

10 (a)	product of density (of medium) and speed of sound (in the medium)	B1	[1]
(b)	α would be nearly equal to 1	M1	
	<i>either</i> reflected intensity would be nearly equal to incident intensity		
	<i>or</i> coefficient for transmitted intensity = $(1 - \alpha)$	M1	
	transmitted intensity would be small	A1	[3]
(c) (i)	$\alpha = (1.7 - 1.3)^2 / (1.7 + 1.3)^2$	C1	
	= 0.018	A1	[2]
(ii)	attenuation in fat = $\exp(-48 \times 2x \times 10^{-2})$	C1	
	$0.012 = 0.018 \exp(-48 \times 2x \times 10^{-2})$	C1	
	$x = 0.42 \text{ cm}$	A1	[3]